IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

Summary of methodologies

- Data collection: SpaceX REST API and web scraping from Wikipedia (BeautifulSoup)
- Data wrangling: filtered to Falcon 9, handled missing values, created a binary landing label
- EDA: visualization (scatter, bar, line charts) and SQL queries
- Interactive analytics: Folium maps and a Plotly Dash dashboard
- Predictive analysis: four classification models tuned with GridSearchCV and compared on test data

Executive Summary

- Landing success rose with experience: 43% in flights 1–30 to 83% in flights 61–90
- Differences by launch site, orbit and payload mostly disappear when flights from the same period are compared
- Since flight 31, 47 of 50 attempted landings succeeded; 12 of the 21 flights with no landing attempt were GTO
- Logistic Regression and SVM predicted 15 of 18 test flights (0.833), above the 0.667 baseline; all their errors were false positives
- Main limits: a mixed label (no attempt counted as failure) and a small test set (one flight = 5.6 points)
- Recommendation: when bidding, first check whether SpaceX plans to recover the booster

Introduction

Project background and context

- SpaceX advertises Falcon 9 launches at about $62 million, while other providers charge $165 million or more (prices stated in the course brief).
- Much of the saving comes from recovering and reusing the first stage.
- If we can predict whether the first stage will land, we can estimate the cost of a launch.
- As a data scientist at Space Y, a new rocket company, I use public launch data and machine learning (not rocket science) to support competitive bidding against SpaceX.

Introduction

Problems you want to find answers

- Can we predict whether the Falcon 9 first stage will land successfully?
- Which factors are related to landing success: payload mass, orbit type, launch site, flight number, booster version?
- Where are the launch sites, and what is near them (coastline, highways, railways)?
- Which classification model performs best, and how reliable is the result with a small dataset?

Section 1

Methodology



Methodology

Data collection methodology

- Collected launch data from the SpaceX REST API (with a course-provided snapshot when the API timed out)
- Notebook: 01 Data collection (API) (nbviewer)
- Scraped Falcon 9 launch records from Wikipedia with BeautifulSoup
- Notebook: 02 Web scraping (nbviewer)

Data wrangling

- Filtered to Falcon 9 launches and handled missing values (PayloadMass filled with the mean)
- Converted landing outcomes into a binary label (Class: 1 = landed, 0 = did not land)
- One-hot encoded categorical features (Orbit, LaunchSite, LandingPad, Serial)
- Notebook: 03 Data wrangling (nbviewer)

Methodology

EDA with visualization and SQL

- Scatter, bar and line charts to explore flight number, payload, orbit, launch site and yearly trend
- SQL queries to summarize launch sites, payloads and landing outcomes
- Notebook: 04 EDA with SQL (nbviewer)
- Notebook: 05 EDA with visualization (nbviewer)

Interactive visual analytics with Folium and Plotly Dash

- Folium: launch site markers, color-coded outcomes, distances to nearby coastline, highways and railways
- Plotly Dash: site dropdown, success pie chart, payload range slider and scatter plot
- Notebook: 06 Interactive map (Folium) (nbviewer)
- App files: 07 Dashboard (Dash) (GitHub)
- Notebook: 07 Dashboard check (nbviewer)

Methodology

Predictive analysis with classification models

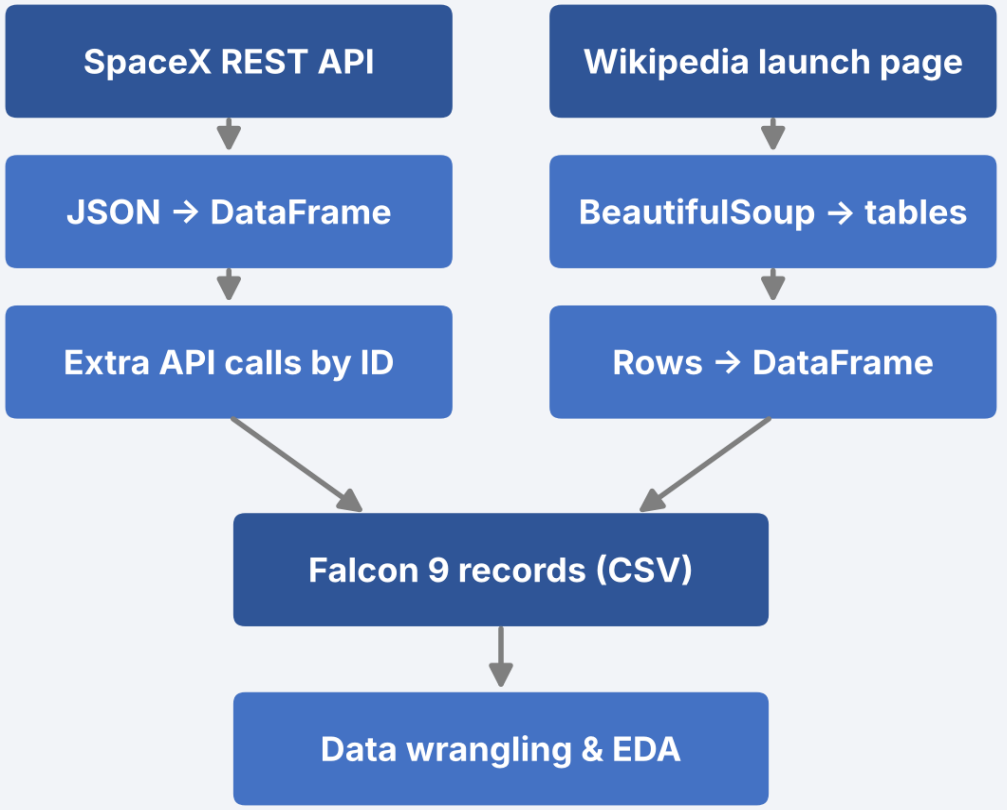
- Standardized features and split data into training and test sets (test: 18 flights)
- Tuned Logistic Regression, SVM, Decision Tree and KNN with GridSearchCV (10-fold cross-validation)
- Evaluated with test accuracy and confusion matrices, compared against a baseline
- Notebook: 08 Machine learning prediction (nbviewer)

Data Collection

Two data sources:

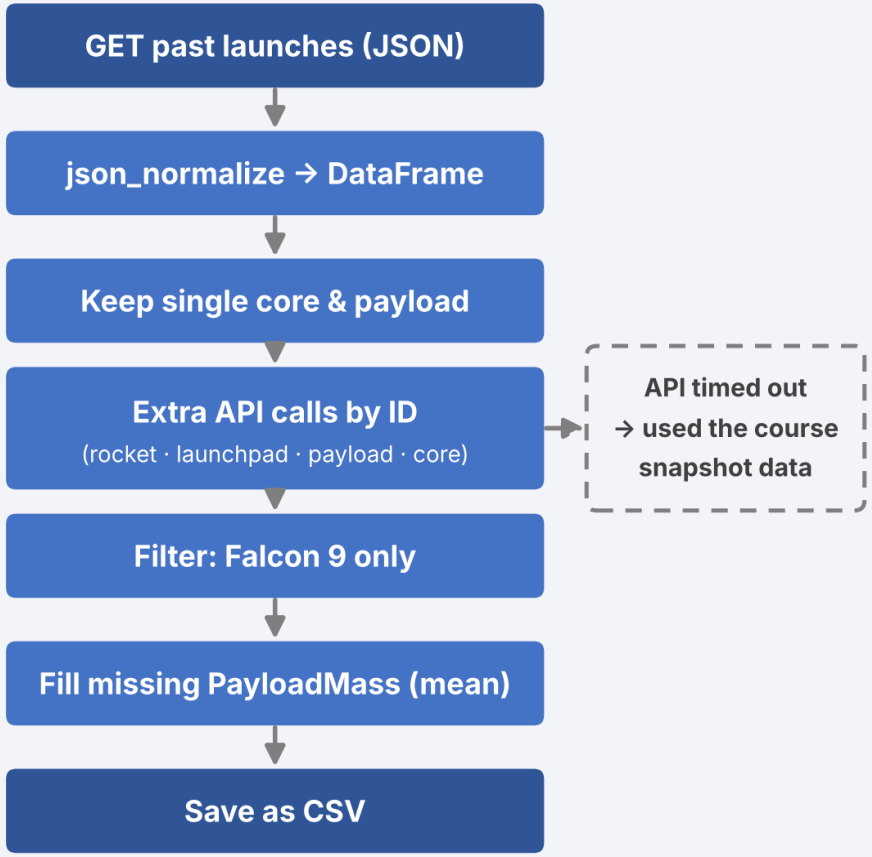
- SpaceX REST API: launch, rocket, payload, launch pad and core data
- Wikipedia: Falcon 9 launch records (web scraping)

- API: GET request → JSON → normalized into a DataFrame → extra API calls by ID for details
- Web scraping: HTML page → BeautifulSoup → launch tables → DataFrame
- Output: Falcon 9 launch records saved as CSV for data wrangling and EDA



Data Collection – SpaceX API

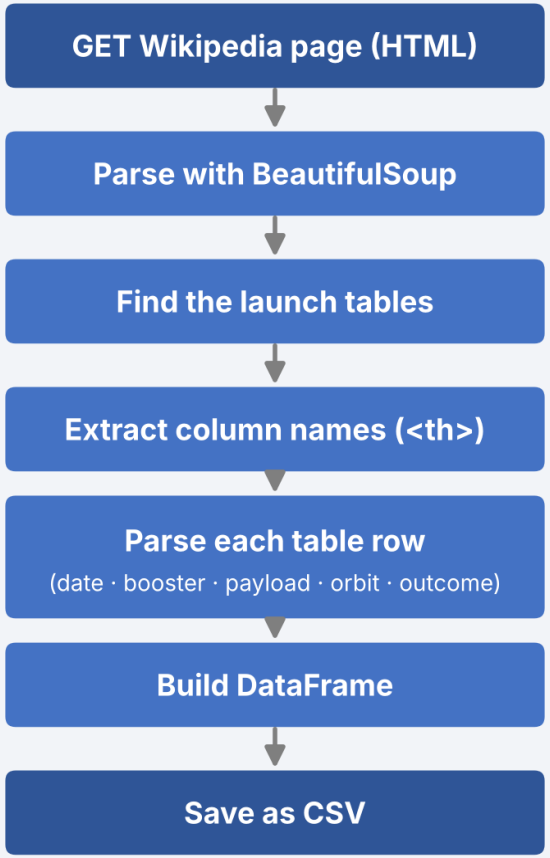
- Requested past launch data from the SpaceX REST API (JSON)
- Converted the JSON to a DataFrame with json_normalize
- Kept launches with a single core and a single payload
- Used the IDs to call the API again for rocket, launchpad, payload and core details
- Kept Falcon 9 launches only
- Filled missing PayloadMass values with the mean
- Saved the cleaned data as a CSV file



GitHub: https://github.com/hokuamadata/spacex-falcon9-landing-prediction/blob/main/01_data_collection_api.ipynb

Data Collection - Scraping

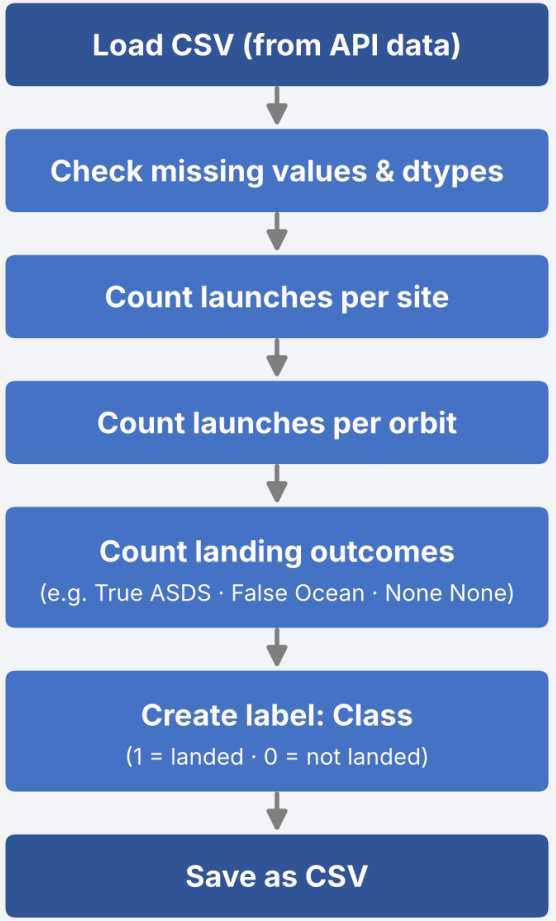
- Requested the Wikipedia page “List of Falcon 9 and Falcon Heavy launches” with requests (HTML)
- Parsed the HTML with BeautifulSoup
- Found the launch tables with find_all("table")
- Extracted column names from the table header cells (<th>)
- Parsed each table row into a dictionary (date, booster, payload, orbit, outcome, etc.)
- Converted the dictionary to a DataFrame
- Saved the data as a CSV file



GitHub: https://github.com/hokuamadata/spacex-falcon9-landing-prediction/blob/main/02_web_scraping.ipynb

Data Wrangling

- Loaded the CSV file created from the SpaceX API data
- Checked the percentage of missing values and the data type of each column
- Counted the launches at each launch site (value_counts)
- Counted the launches to each orbit type
- Counted the landing outcomes (e.g. True ASDS = landed on a drone ship, False Ocean = failed ocean landing)
- Grouped the unsuccessful outcomes i nto a set (bad_outcomes)
- Created the label column Class (1 = landed, 0 = not landed) to use as the target for prediction
- Saved the data as a CSV file for the next steps (EDA and modeling)



GitHub: https://github.com/hokuamadata/spacex-falcon9-landing-prediction/blob/main/03_data_wrangling.ipynb

EDA with SQL

Launch sites

DISTINCT · LIKE · LIMIT

- Listed the unique launch site names
- Displayed 5 records where the site name begins with "CCA"

Payload mass

SUM · AVG · subquery (MAX)

- Total payload mass carried for NASA (CRS)
- Average payload mass of booster version F9 v1.1
- Booster versions that carried the maximum payload mass

Successful landings

MIN · BETWEEN

- Date of the first successful landing on a ground pad
- Boosters that landed on a drone ship with a payload of 4,000–6,000 kg

Outcome counts

COUNT · GROUP BY · ORDER BY · substr

- Number of successful and failed mission outcomes
- Failed drone ship landings in 2015, by month
- Landing outcomes ranked by count (2010-06-04 to 2017-03-20)

GitHub: https://github.com/hokuamadata/spacex-falcon9-landing-prediction/blob/main/04_eda_sql_sqllite.ipynb

EDA with Data Visualization

Scatter plots

Why: to see how two variables relate to landing success

- Flight Number vs. Payload Mass / Launch Site / Orbit Type
- Payload Mass vs. Launch Site / Orbit Type

Bar chart

Why: to compare success rates across categories

- Success rate by Orbit Type

Line chart

Why: to show how the success rate changed over time

- Yearly average success rate

Feature engineering

Why: to prepare the data for machine learning

- One-hot encoding of categorical columns (get_dummies)
- Converted all columns to float64 and saved as CSV

GitHub: https://github.com/hokuamadata/spacex-falcon9-landing-prediction/blob/main/05_eda_with_visualization.ipynb

Build an Interactive Map with Folium

Circles & labels at launch sites

folium.Circle · Marker with DivIcon

- To show where each launch site is on the map and label it by name

Launch markers in a cluster

MarkerCluster · green = success, red = failure

- Many launches share the same coordinates, so clustering keeps the map readable
- Shows at a glance which sites have higher success rates

Mouse position display

MousePosition

- To read the coordinates of nearby places (coastline, railway, highway, city) by pointing at them

Lines & distance labels

PolyLine · Marker with DivIcon

- To measure how close launch sites are to the coastline, railways, highways and cities
- To see what affects where launch sites are built

GitHub: https://github.com/hokuamadata/spacex-falcon9-landing-prediction/blob/main/06_interactive_map_folium.ipynb

Build a Dashboard with Plotly Dash

Launch site dropdown

All Sites or 1 of 4 sites

- Switches the charts between all sites and a single site, so the sites can be compared

Success pie chart

Changes with the selected site

- All Sites: each site's share of all successful launches, to see which site has the most successes
- One site: the ratio of successes to failures at that site, to see its success rate

Payload range slider

0 – 10,000 kg

- Narrows down the payload range, to see how payload mass relates to success

Payload vs. outcome scatter plot

Colored by booster version

- Shows which payload ranges and booster versions have higher success rates

GitHub: https://github.com/hokuamadata/spacex-falcon9-landing-prediction/tree/main/07_interactive_dashboard_dash

Predictive Analysis (Classification)

Build

- Used Class as the target (Y) and the one-hot encoded features as X
- Standardized X with StandardScaler
- Split the data into training (80%) and test (20%) sets with train_test_split

Improve

- Tuned the hyperparameters of 4 models with GridSearchCV (10-fold cross-validation)
  - Logistic Regression, SVM, Decision Tree, KNN

Evaluate

- Checked the best parameters and the cross-validation accuracy (best_params_, best_score_)
- Calculated the accuracy on the test data (score)
- Plotted the confusion matrix to see the types of errors (e.g. false positives)

Find the best model

- Compared the test accuracy of the 4 models and selected the best one

GitHub: https://github.com/hokuamadata/spacex-falcon9-landing-prediction/blob/main/08_machine_learning_prediction.ipynb

Results

EDA results

Charts & SQL

- Success rate rose overall, especially after 2015
- The 2018 dip came from flights with no landing attempt (attempted landings: about 92%)
- SSO: 100% over 5 flights; VLEO: about 86% (12 of 14)
- ES-L1, GEO, HEO: 100%, but only 1 flight each
- KSC LC-39A had the most successful launches (41.7%)

Interactive analytics

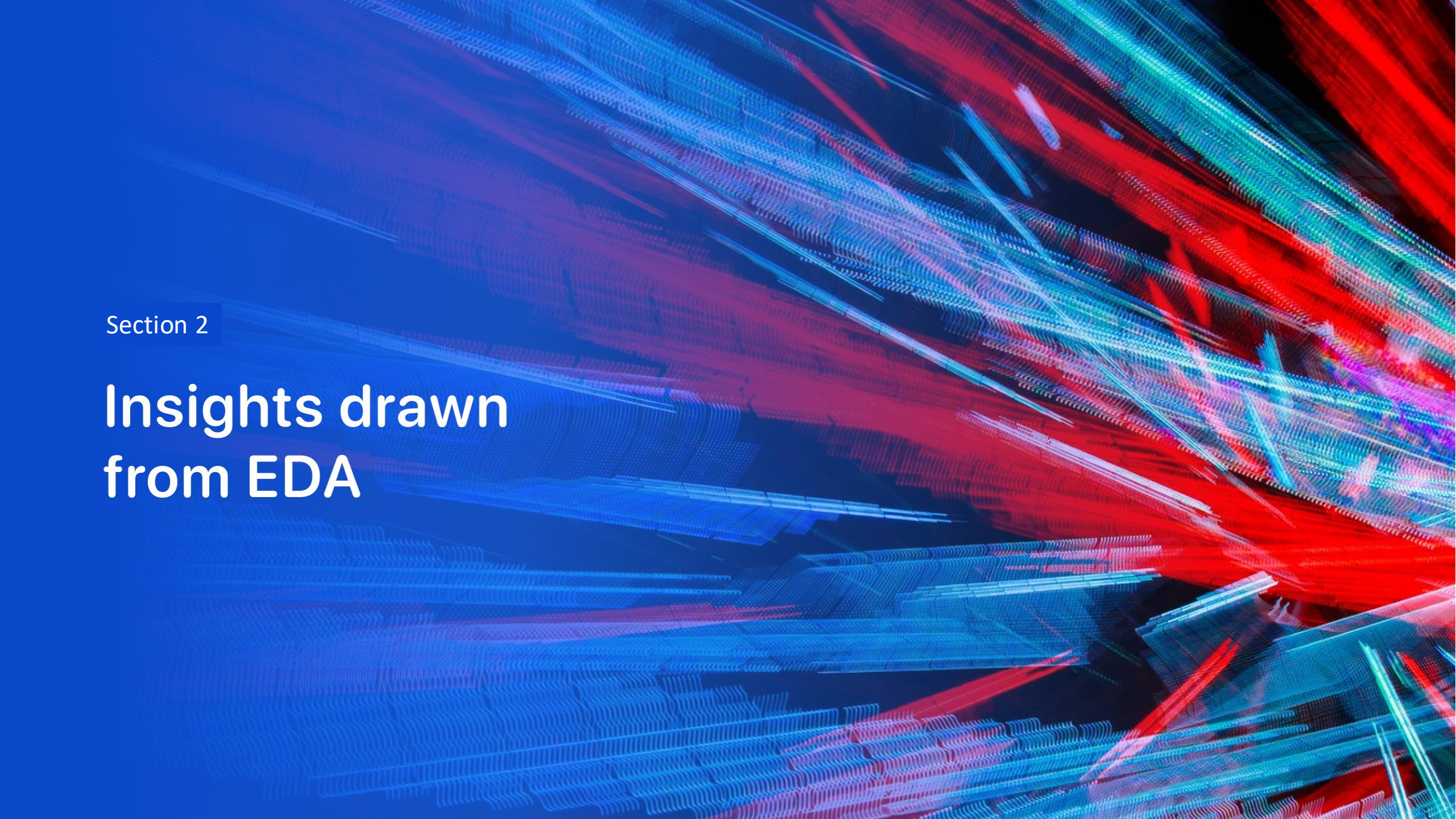
Folium map · Dash dashboard

- Folium: launch sites are close to the coastline and railways, and far from cities and highways
- Dash: payloads of 2,000–4,000 kg had the highest success rate (61.9%, 13 of 21)

Predictive analysis

4 classification models

- Logistic Regression and SVM tied for the best test accuracy: 0.83 (15 of 18)
- Main error type for both: false positives (predicted a landing that failed)


Section 2
Insights drawn from EDA

Two of the four launch site names likely refer to the same pad

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Observations

- SELECT DISTINCT returned 4 unique launch sites: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A and CCAFS SLC-40.

Insight

2010–16 — **CCAFS LC-40**
Jun 2010 – Aug 2016

2017–20 — **CCAFS SLC-40**
Dec 2017 – Nov 2020

The periods never overlap, so they are likely the same pad under two names. The EDA section treats them as one site.

“No attempt” was the most common landing outcome before 2017

```
%%sql
SELECT Landing_Outcome, COUNT(*) AS Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Count DESC
```

```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Observations

- The most common landing outcome was “No attempt” (10 flights), followed by Success (drone ship) and Failure (drone ship) (5 each).
- GROUP BY counted each outcome, and ORDER BY … DESC ranked them.

Insight

10 **No attempt**
out of 31 flights

8 **Landed successfully**
out of 31 flights

Class = 0 often means no landing attempt, not a failed landing.

The first five CCA records are the earliest flights (2010–2013)

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Observations

- LIKE 'CCA%' selected launch sites that begin with “CCA”, and LIMIT 5 showed the first 5 records.
- All 5 records are from CCAFS LC-40 between 2010 and 2013, with small payloads (0–677 kg) and no successful landings.
- Without ORDER BY, LIMIT 5 returns rows in table order, which happens to be by date here.

The NASA (CRS) total depends on how the customer is matched

Partial match (LIKE)

```
%%sql SELECT SUM(PAYLOAD_MASS__KG_)
    AS sum_payload FROM SPACEXTABLE
    WHERE Customer LIKE '%NASA (CRS)%'
```

```
 * sqlite:///my_data1.db
Done.
```

sum_payload
48213

Exact match (=)

```
%%sql SELECT SUM(PAYLOAD_MASS__KG_)
    AS sum_payload FROM SPACEXTABLE
    WHERE Customer = 'NASA (CRS)'
```

```
 * sqlite:///my_data1.db
Done.
```

sum_payload
45596

Observations

- SUM(PAYLOAD_MASS__KG_) with WHERE Customer LIKE '%NASA (CRS)%' returned a total payload of 48,213 kg.
- This includes one shared mission (NASA (CRS), Kacific 1, 2,617 kg); an exact match (= 'NASA (CRS)') gives 45,596 kg.

Including numbered boosters changes the F9 v1.1 average

Partial match (LIKE)

```
%%sql
SELECT ROUND(AVG(PAYLOAD_MASS__KG_), 1)
AS avg_payload, COUNT(*) AS flights
FROM SPACEXTABLE
WHERE Booster_Version LIKE '%F9 v1.1%'
```

```
 * sqlite:///my_data1.db
Done.
```

avg_payload	flights
2534.7	15

Exact match (=)

```
%%sql
SELECT AVG(PAYLOAD_MASS__KG_)
AS avg_payload, COUNT(*) AS flights
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1'
```

```
 * sqlite:///my_data1.db
Done.
```

avg_payload	flights
2928.4	5

Observations

- Including numbered boosters (LIKE '%F9 v1.1%', e.g. F9 v1.1 B1010), the average is about 2,534.7 kg (15 flights).
- An exact match (Booster_Version = 'F9 v1.1') gives 2,928.4 kg (5 flights).

The first successful landing came in December 2015, on a ground pad

```
%%sql
SELECT MIN(Date) AS first_success_date
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)'
```

```
 * sqlite:///my_data1.db
Done.
```

first_success_date
2015-12-22

Observations

- MIN(Date) with WHERE Landing_Outcome = 'Success (ground pad)' returned 2015-12-22.
- Because the dates are stored as YYYY-MM-DD text, MIN returns the earliest date.

Insight

2015-12-22

First successful landing

of any kind (on a ground pad)

33% → 83%

Yearly success rate

2015 → 2017

After this first success, the yearly success rate rose sharply.

The next section (EDA with visualization) shows this trend.

Later drone ship landings are hidden by missing labels

```
%%sql SELECT Booster_Version, PAYLOAD_MASS__KG_, Landing_Outcome
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
AND PAYLOAD_MASS__KG_ > 4000
AND PAYLOAD_MASS__KG_ < 6000
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version	PAYLOAD_MASS__KG_	Landing_Outcome
F9 FT B1022	4696	Success (drone ship)
F9 FT B1026	4600	Success (drone ship)
F9 FT B1021.2	5300	Success (drone ship)
F9 FT B1031.2	5200	Success (drone ship)

Observations

- 4 boosters landed on a drone ship with a payload of 4,000–6,000 kg: F9 FT B1022, F9 FT B1026,
  F9 FT B1021.2 and F9 FT B1031.2.
- All of them flew between 2016 and 2017.

Insight

4 **Boosters in the result**
May 2016 – Oct 2017

7 **Not counted: "Success" without location**
Aug 2018 – Nov 2020, same payload range

After mid-2018, landings are recorded only as "Success",
so later drone ship landings may be missing from this result.

Almost every mission succeeded, but only 60% of boosters landed

Mission outcome

```
%%sql SELECT TRIM( Mission_Outcome ) AS Mission,
COUNT(*) AS Count
FROM SPACEXTABLE GROUP BY TRIM( Mission_Outcome )
```

```
 * sqlite:///my_data1.db
Done.
```

Mission	Count
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

Landing outcome

```
%%sql
SELECT CASE WHEN TRIM(Landing_Outcome) LIKE 'Success%'
            THEN 'Success' ELSE 'Not success' END AS Landing,
       COUNT(*) AS Count
FROM SPACEXTABLE
GROUP BY Landing
```

```
 * sqlite:///my_data1.db
Done.
```

Landing	Count
Not success	40
Success	61

Observations

- Mission outcome: 100 of 101 missions succeeded (including 1 with payload status unclear); 1 failed in flight.
- Landing outcome: 61 of 101 boosters landed successfully.

Insight

100 **Missions succeeded** out of 101 (99%)

61 **Landings succeeded** out of 101 (60%)

Missions almost always succeeded,
but far fewer boosters landed.
This project predicts landing success.

Only reused Block 5 boosters carried the maximum payload

```
%%sql SELECT TRIM( Booster_Version )
AS Booster_Version, PAYLOAD_MASS__KG_
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_)
FROM SPACEXTABLE)
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

Observations

- The subquery found the maximum payload mass (15,600 kg), and 12 launches carried it.
- All of them used Block 5 (F9 B5) boosters.

Insight

12 **Launches with the maximum payload**
15,600 kg each

6 **Physical boosters, all reused**
these were their 2nd to 7th flights (.2 – .7)

The heaviest payloads were carried only by reused Block 5 boosters, so reuse did not prevent the heaviest launches.

Every drone ship landing in 2015 failed

```
%%sql SELECT substr(Date, 6, 2) AS month,
Landing_Outcome, Booster_Version, Launch_Site
FROM SPACEXTABLE WHERE substr(Date, 1, 4) = '2015'
AND Landing_Outcome LIKE '%Failure (drone ship)%'
```

```
 * sqlite:///my_data1.db
Done.
```

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Observations

- In 2015, 2 drone ship landings failed: in January (F9 v1.1 B1012) and April (F9 v1.1 B1015), both from CCAFS LC-40.
- substr(Date, 6, 2) extracted the month, and substr(Date, 1, 4) filtered the year.

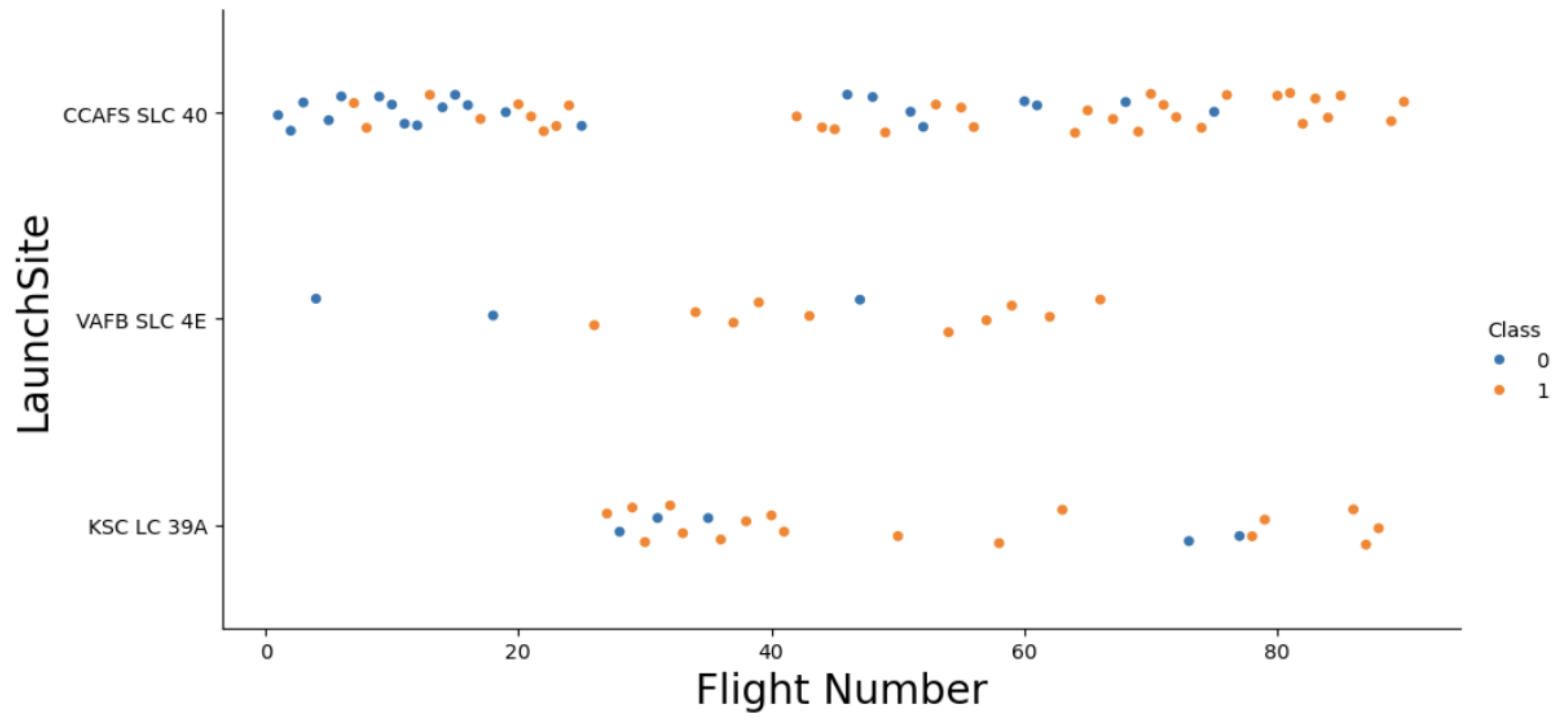
Insight

0 / 2 — **Drone ship landings in 2015**
both attempts failed (Jan and Apr)

Dec — **First successful landing**
2015-12-22, on a ground pad

In 2015, every drone ship attempt failed. The first success came at the end of the year, on a ground pad instead (shown earlier).

Success rate improved as flight number increased



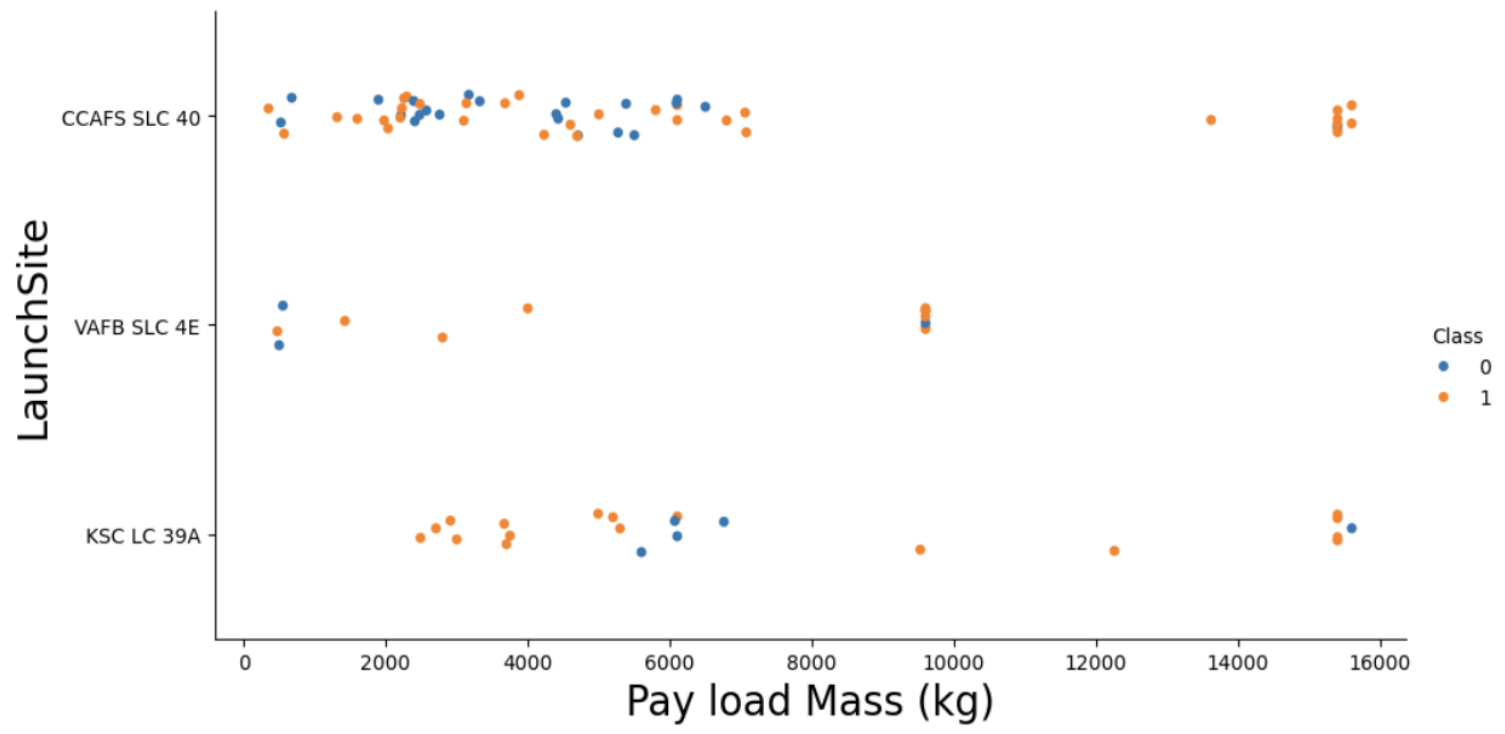
Observations

- Overall, the success rate increased as the flight number increased.
- CCAFS SLC 40 had the most launches, and most of its early launches failed.
- KSC LC 39A was first used from flight 27, so it has fewer early failures than the other two sites.

Insight

The apparent difference in success rates between sites is likely due to timing, not the sites themselves (from flight 27 onward: CCAFS SLC 40 75%, KSC LC 39A 77%).

Payload mass alone does not explain landing success



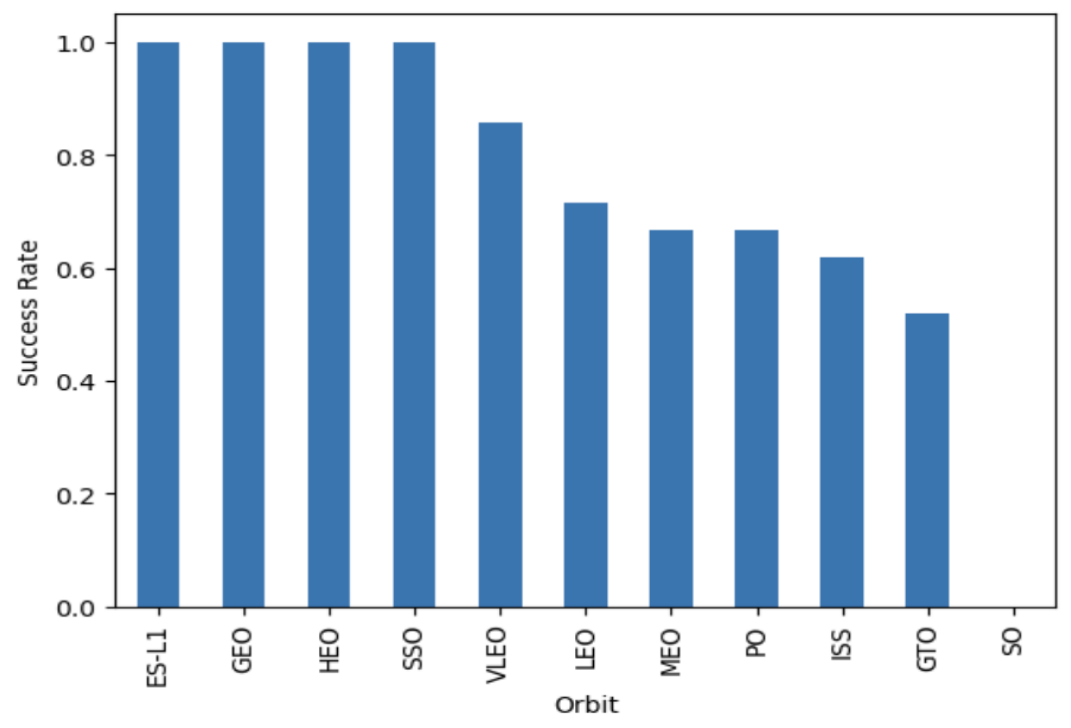
Observations

- VAFB SLC 4E has no launches with a payload mass over 10,000 kg.
- Most launches with heavy payloads (over 10,000 kg) were successful (13 of 15).
- For light payloads, successes and failures are mixed at all sites.

Insight

Heavy payloads (over 10,000 kg) were all launched from flight 63 onward, so their high success rate is likely due to timing, not payload mass (from flight 63: heavy 13 of 15, light 11 of 13).

The highest success rates come from orbits with very few flights



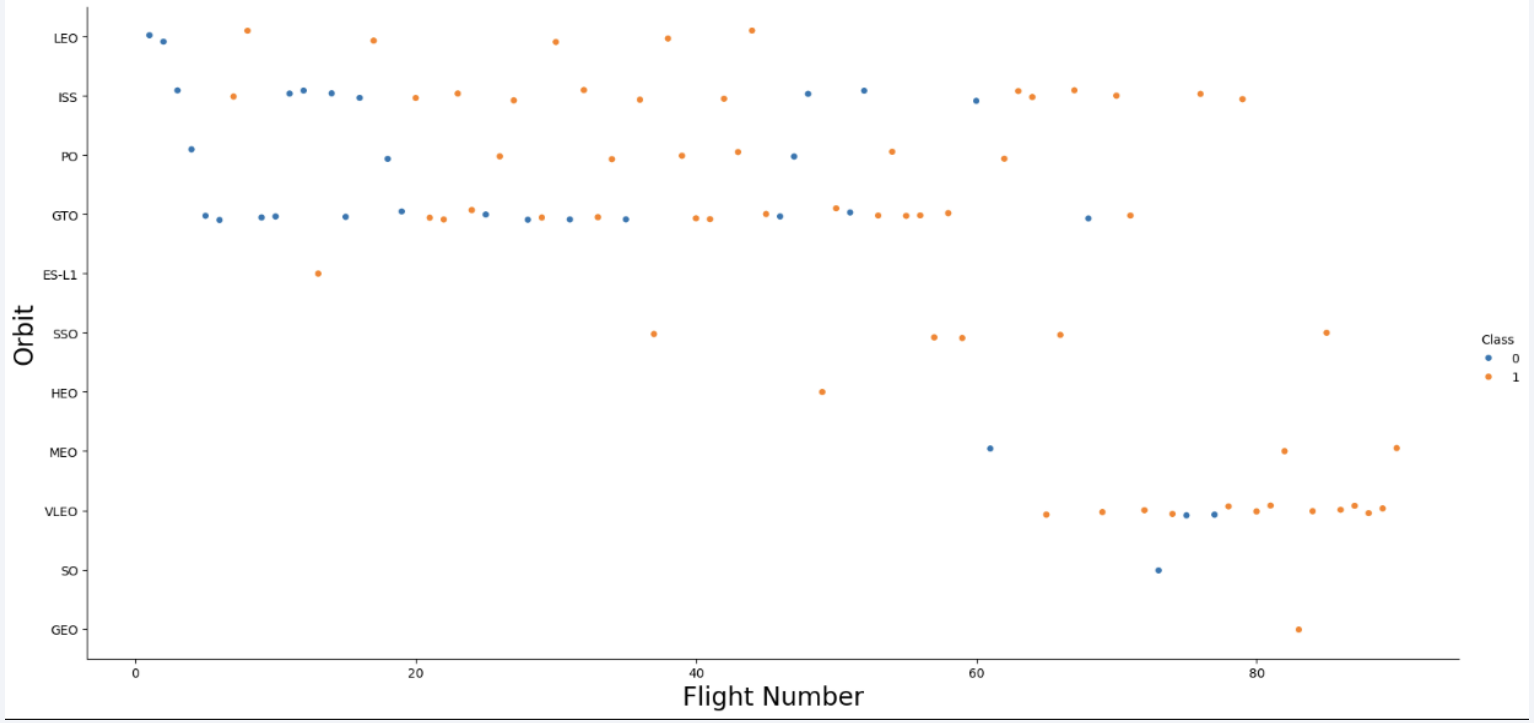
Observations

- ES-L1, GEO and HEO had a 100% success rate, but with only 1 flight each, so this should be interpreted with caution.
- SSO is the only orbit with a 100% success rate over multiple flights (5 of 5).
- VLEO had a high success rate over many flights (about 86%, 12 of 14).

Insight

Orbit type alone does not explain landing success, because the highest rates come from orbits with very few flights. The next slide checks orbit type together with flight number.

Flight Number vs. Orbit Type



Observations

- GTO seems to have many failures, but 12 of its 13 failures were flights with no landing attempt.
- Orbits with few flights (9 or fewer), and even ISS with 21 flights, show no clear relationship between flight number and success.
- VLEO launches are concentrated in the later period (flights 65–89); all 14 attempted a landing, and 12 succeeded (about 86%).

Insight

VLEO's high success rate may also reflect timing, because all of its launches were in the later period (flights 65–89).

Many “failures” in GTO were flights that never tried to land

Orbit	Flights	Attempted	Landed	Rate_attempted
ES-L1	1	1	1	1.00
GEO	1	1	1	1.00
GTO	27	15	14	0.93
HEO	1	1	1	1.00
ISS	21	17	13	0.76
LEO	7	5	5	1.00
MEO	3	2	2	1.00
PO	9	8	6	0.75
SO	1	0	0	NaN
SSO	5	5	5	1.00
VLEO	14	14	12	0.86

*Attempted = Outcome does not start with “None”
*SO: no attempted landings

Insight

52%

All GTO flights

14 of 27 landed

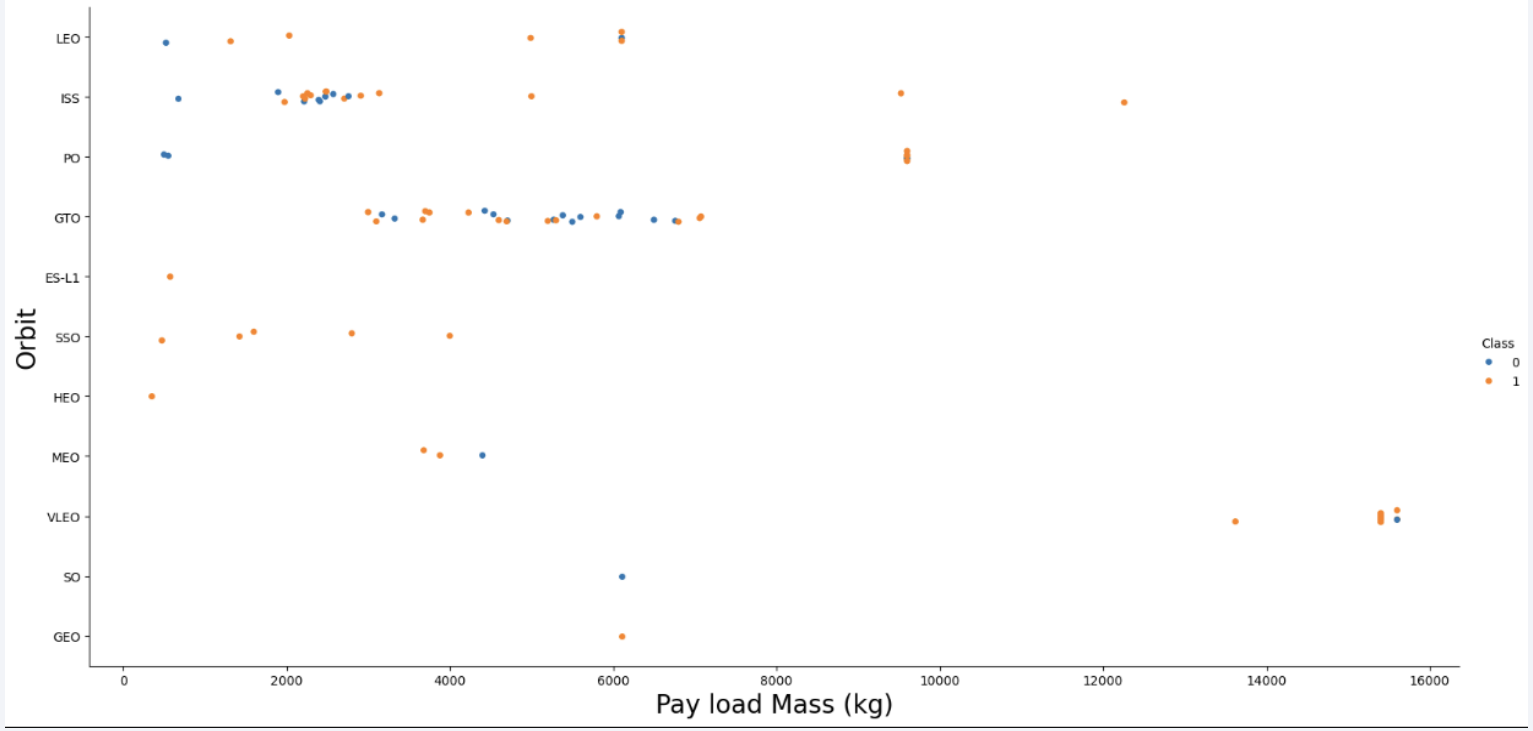
93%

Attempted landings only

14 of 15 landed

GTO's low success rate is mainly due to flights with no landing attempt.
Class = 0 mixes landing failures with flights that never tried to land.

Every payload over 13,000 kg went to VLEO



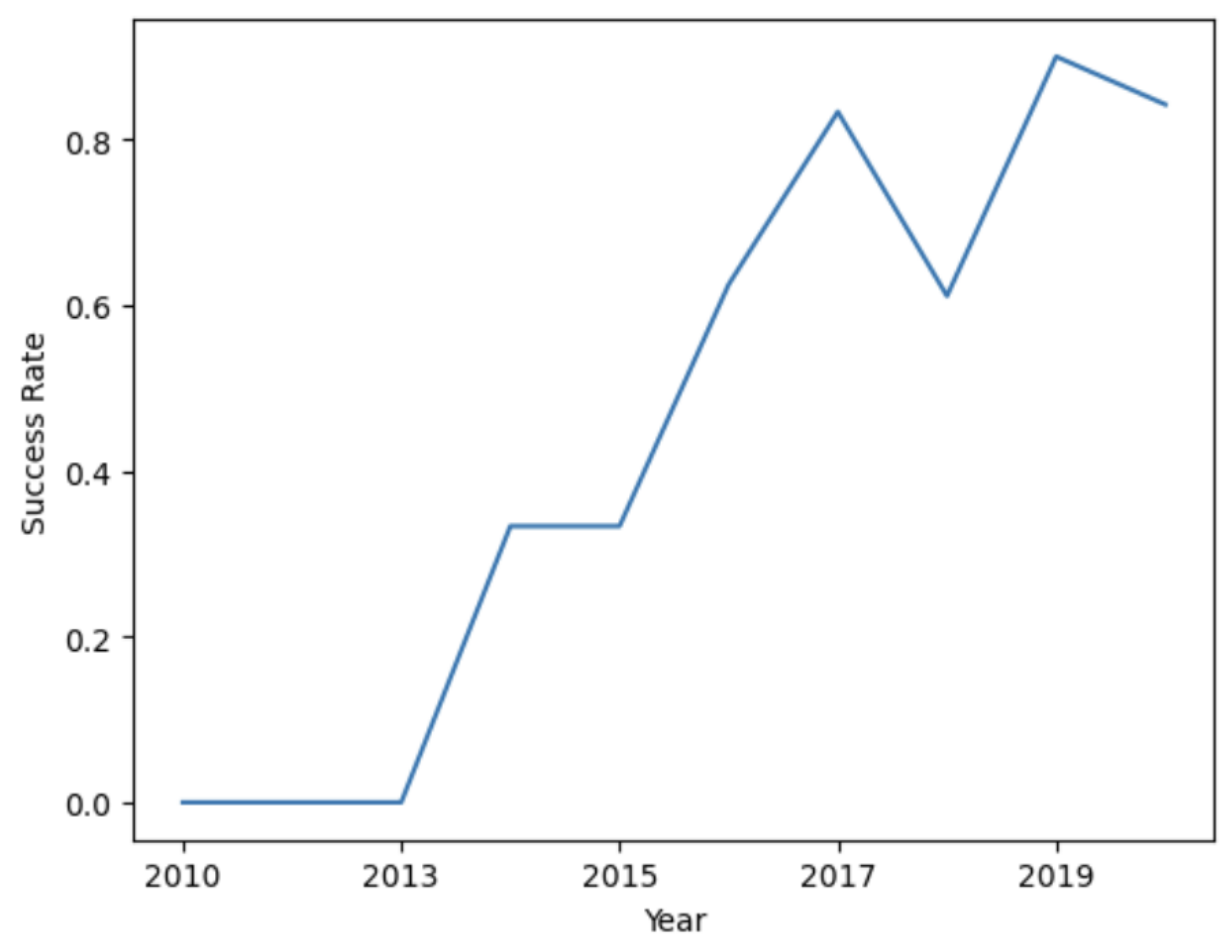
Observations

- All launches with a payload mass over 13,000 kg went to VLEO.
- For these heavy payloads, most VLEO landings were successful (12 of 14).
- GTO launches were concentrated between about 3,000 and 7,000 kg, with successes and failures mixed across that range.

Insight

Payload mass, orbit and timing overlap: every launch over 13,000 kg went to VLEO in the later period, so its high success rate cannot be attributed to any one of these factors alone.

The success rate rose sharply from 2015 to 2017



Observations

- The success rate was 0% until 2013, then rose sharply from 2015 to 2017.
- There is a dip in 2018, but the rate recovered to about 90% in 2019 (see next slide).

The 2018 dip came from flights with no landing attempt

Date	Flights	Attempted	Landed	Rate_attempted
2010	1	0	0	NaN
2012	1	0	0	NaN
2013	3	1	0	0.00
2014	6	3	2	0.67
2015	6	4	2	0.50
2016	8	7	5	0.71
2017	18	15	15	1.00
2018	18	12	11	0.92
2019	10	9	9	1.00
2020	19	18	16	0.89

*Attempted = Outcome does not start with “None”
*2010, 2012: no attempted landings

Insight

61%

All 2018 flights

11 of 18 landed

92%

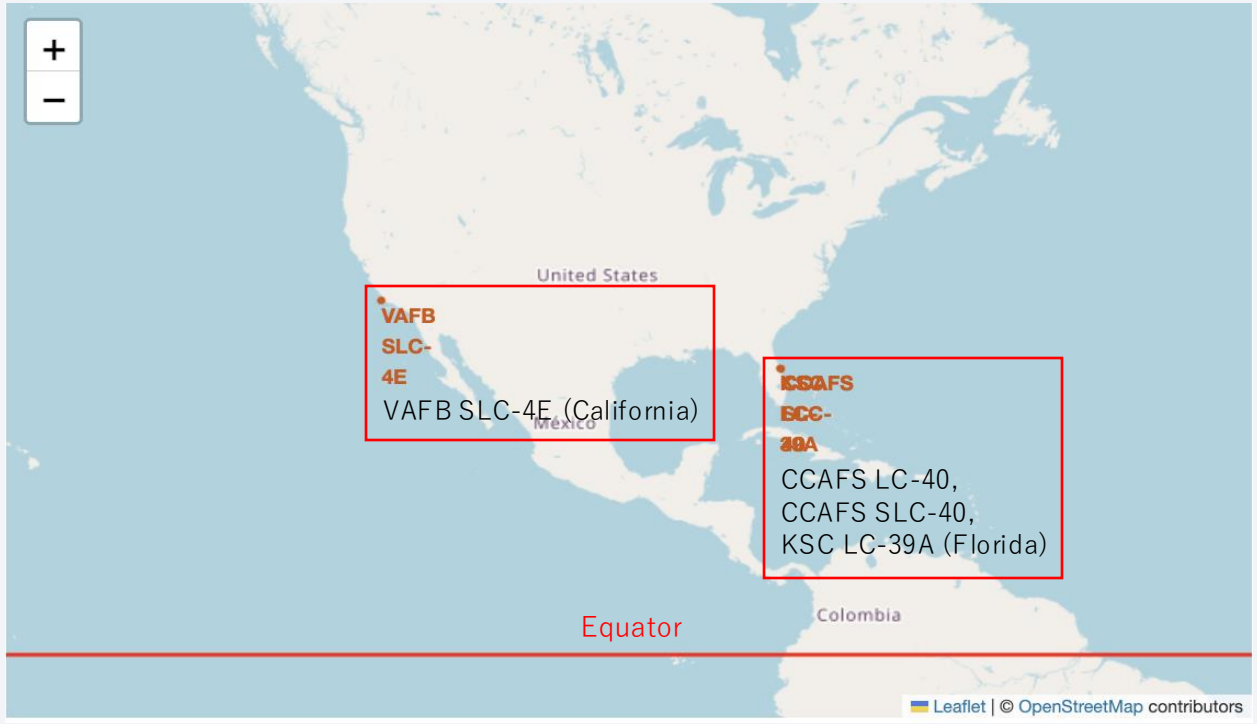
Attempted landings only

11 of 12 landed

The 2018 dip is mainly due to flights with no landing attempt (6 of 18), the same Class = 0 issue as GTO.


Section 3
Launch Sites
Proximities Analysis

All launch sites are on the US coasts, in the south of the country



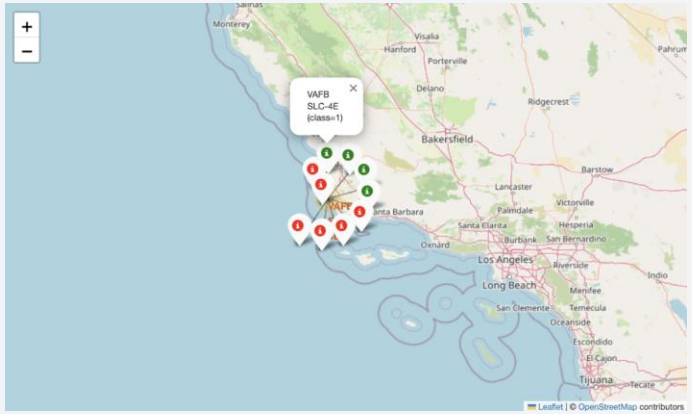
Observations

- The launch sites are in two areas: 1 on the west coast (California) and 3 on the east coast (Florida).
- All launch sites are right on the coast.
- All launch sites are in the southern part of the US, but still far from the equator: about 28.5° N (Florida) and 34.6° N (California).

Insight

Being in the south helps eastward launches from Florida, which can use more of the Earth's rotation. VAFB is the exception: it launches south into polar orbits over the Pacific.

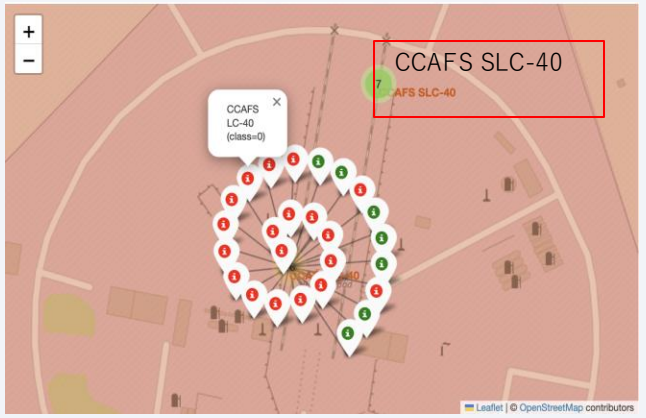
KSC LC-39A had the most successful landings, mainly because it was used later



VAFB SLC-4E: 4 of 10 green



KSC LC-39A: 10 of 13 green



CCAFS LC-40: 7 of 26 green

Observations

- Green = successful landing, red = failure; each cluster groups launches from one site.
- KSC LC-39A has mostly green markers (10 of 13), while CCAFS LC-40 has the most launches but only 7 of 26 green.
- The CCAFS LC-40 cluster sits right next to CCAFS SLC-40 (about 100 m apart).

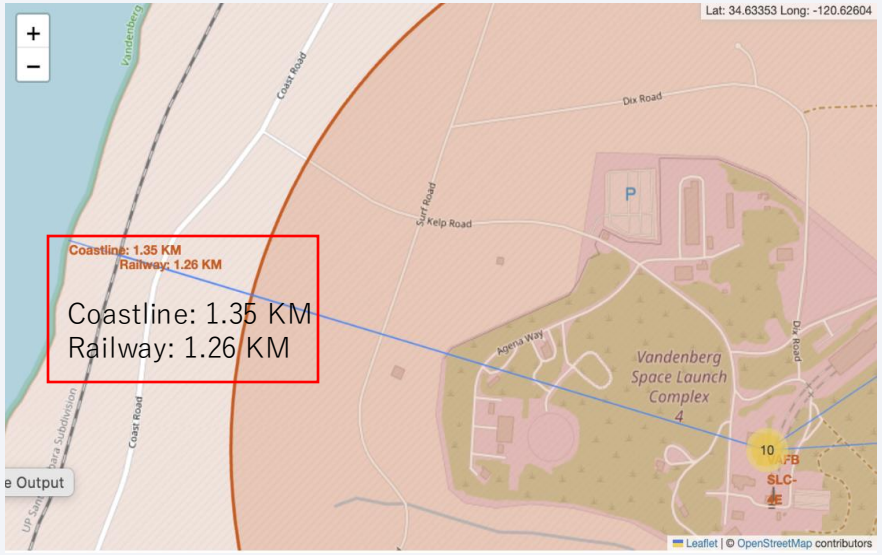
Insight

100 m — **Between CCAFS LC-40 and SLC-40**
likely the same pad under two names

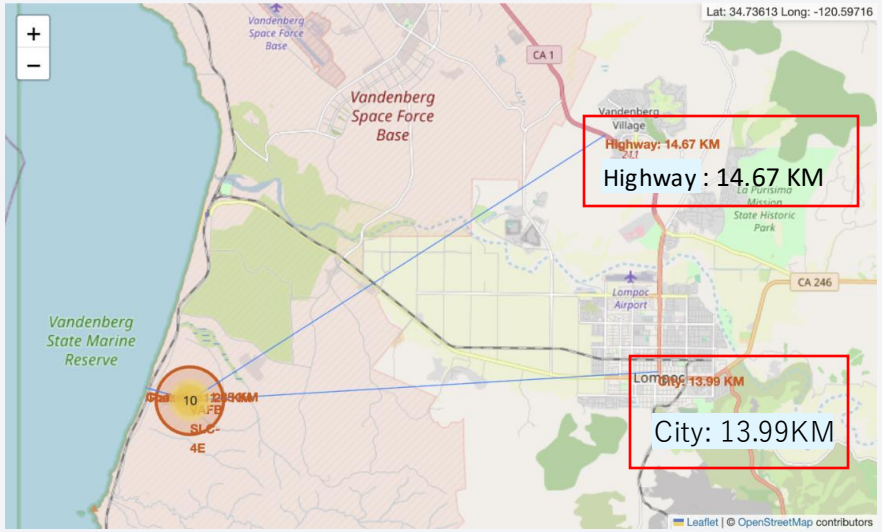
10 / 13 — **KSC LC-39A landings succeeded**
used only from 2017, a later period

KSC has the most green markers, but mainly because it was used later, not because the site itself is better (as in the EDA section).

VAFB SLC-4E is close to the coast and railway, but far from the city



Coastline and railway (zoomed in)



City and highway (zoomed out)

Observations

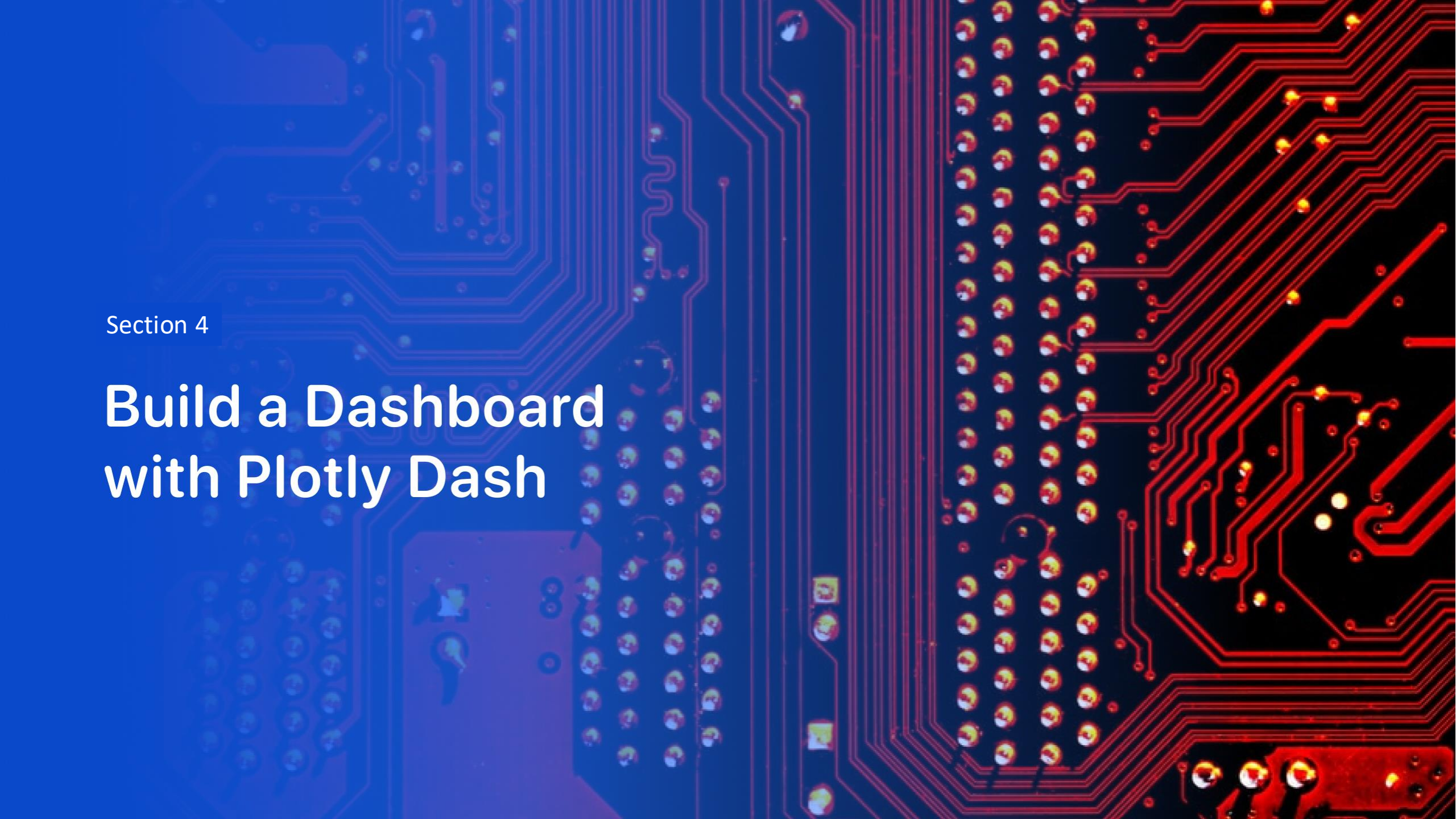
- VAFB SLC-4E is about 1.35 km from the coastline and 1.26 km from the railway (UP Santa Barbara Subdivision).
- The nearest city (Lompoc) is about 13.99 km away, and the measured point on the highway (Cabrillo Highway) is about 14.67 km away.
- Distances are straight-line distances calculated from the coordinates.

Insight

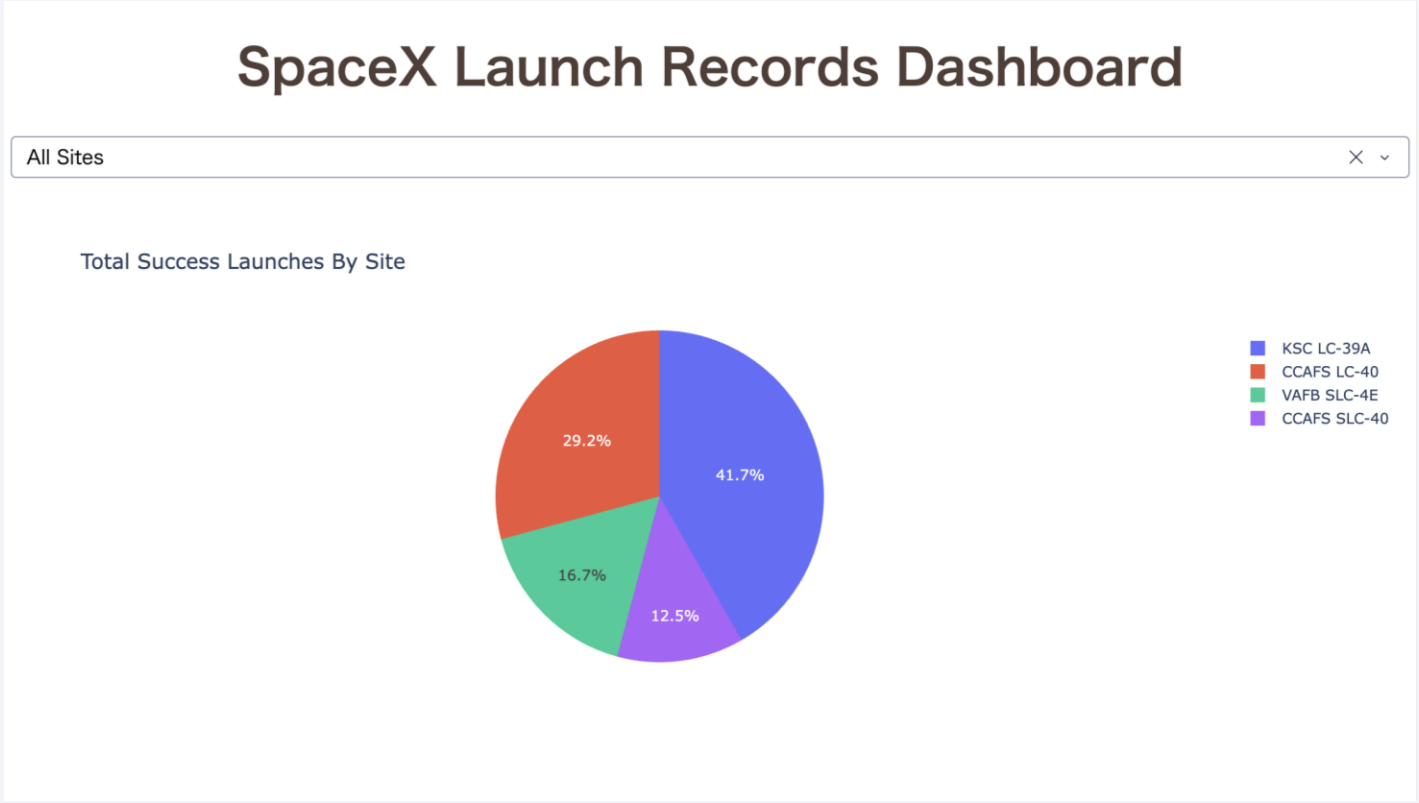
~1 km — **Coastline and railway**
1.35 km and 1.26 km

~14 km — **Nearest city and highway**
Lompoc 13.99 km, Cabrillo Hwy 14.67 km

Close to the coast (launch over the ocean) and railway (heavy transport), but far from the city for safety. Based on one site only.


Section 4
Build a Dashboard with Plotly Dash

KSC LC-39A had the largest share of successful launches



Observations

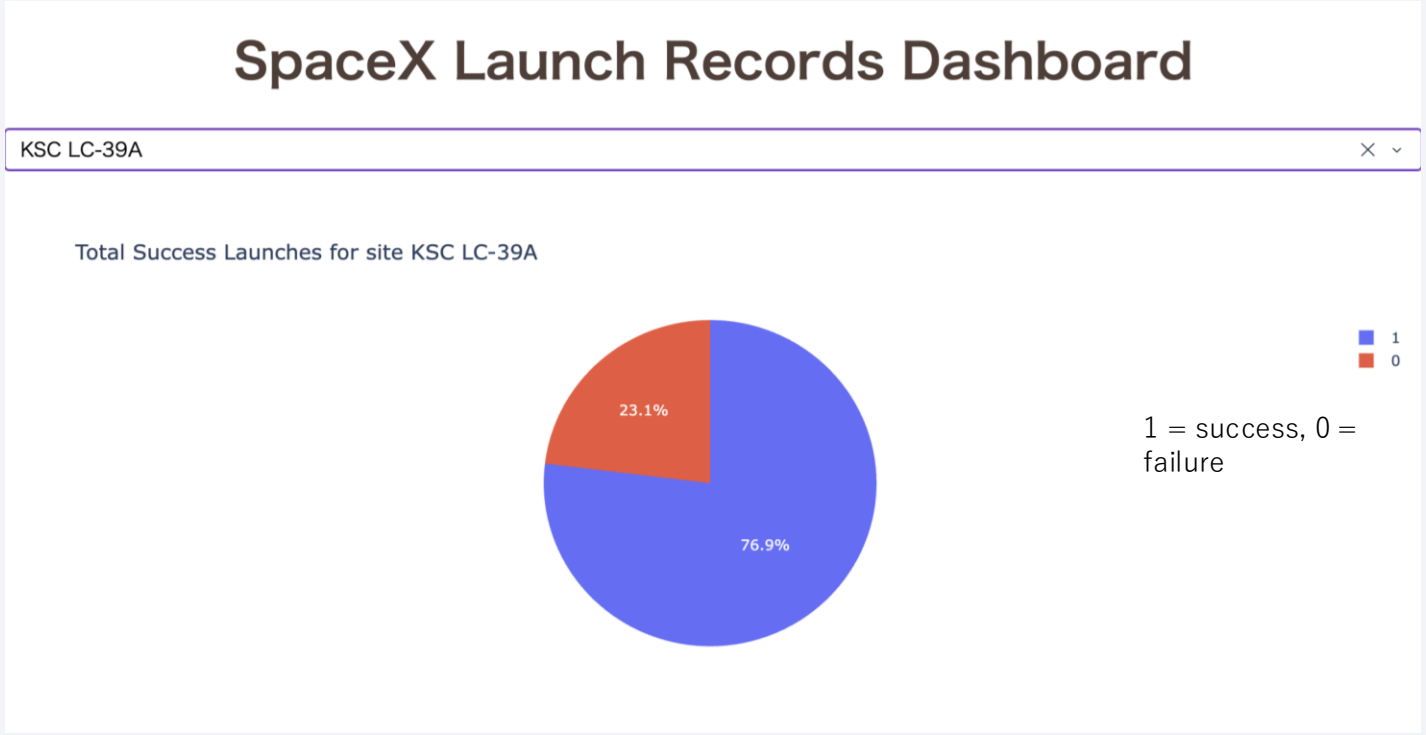
- The pie chart shows each site's share of all 24 successful launches.
- KSC LC-39A had the largest share (41.7%), followed by CCAFS LC-40 (29.2%).
- CCAFS SLC-40 had the smallest share (12.5%).

Insight

41.7%	**KSC LC-39A** 10 of 24 successful launches
41.7%	**CCAFS LC-40 + SLC-40** 7 + 3 = 10, likely the same pad

Counted as one pad, CCAFS ties with KSC. The pie shows shares of successes, which also depend on how many launches each site had.

KSC LC-39A had the highest success rate (76.9%), mainly because it started later



Observations

- When KSC LC-39A is selected, the pie chart shows its success rate: 76.9% success (1) and 23.1% failure (0).
- KSC LC-39A has the highest success rate of all sites (10 of 13 launches).

Insight

76.9% **KSC LC-39A**
10 of 13 launches, used from 2017

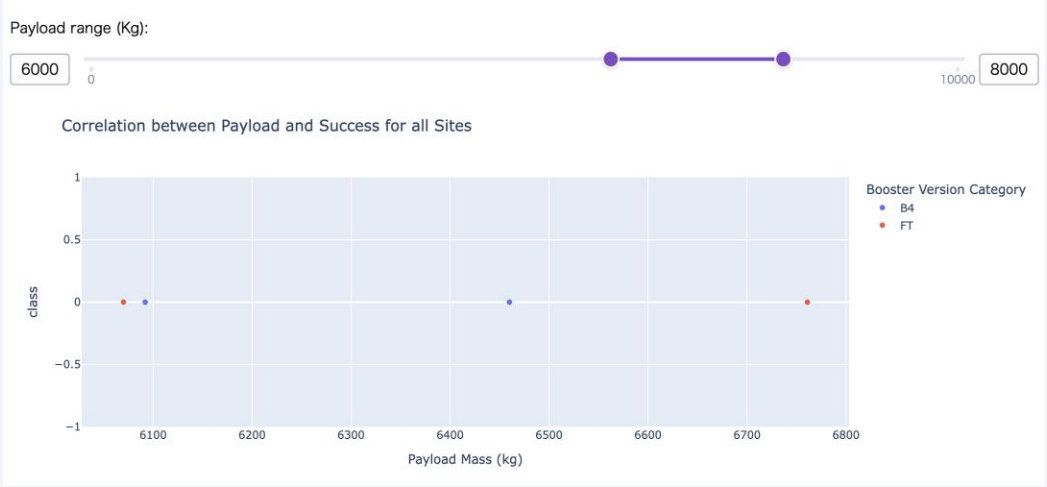
30.3% **CCAFS LC-40 + SLC-40**
10 of 33 launches, used from 2010

Same number of successes, very different rates. KSC's high rate mainly reflects its later start, when landings were more reliable.

2,000–4,000 kg had the highest landing success rate



2,000–4,000 kg: 13 of 21



6,000–8,000 kg: 0 of 4

Observations

- The range slider shows how the landing outcome changes with payload mass.
- 2,000–4,000 kg had the highest success rate (61.9%, 13 of 21).
- 6,000–8,000 kg had the lowest (0 of 4), but all 4 flights made no landing attempt. Above 8,000 kg, 3 of 5 landed.

Insight

61.9% — **2,000–4,000 kg (highest)**
13 of 21 landed

0% — **6,000–8,000 kg (lowest)**
0 of 4, all with no landing attempt

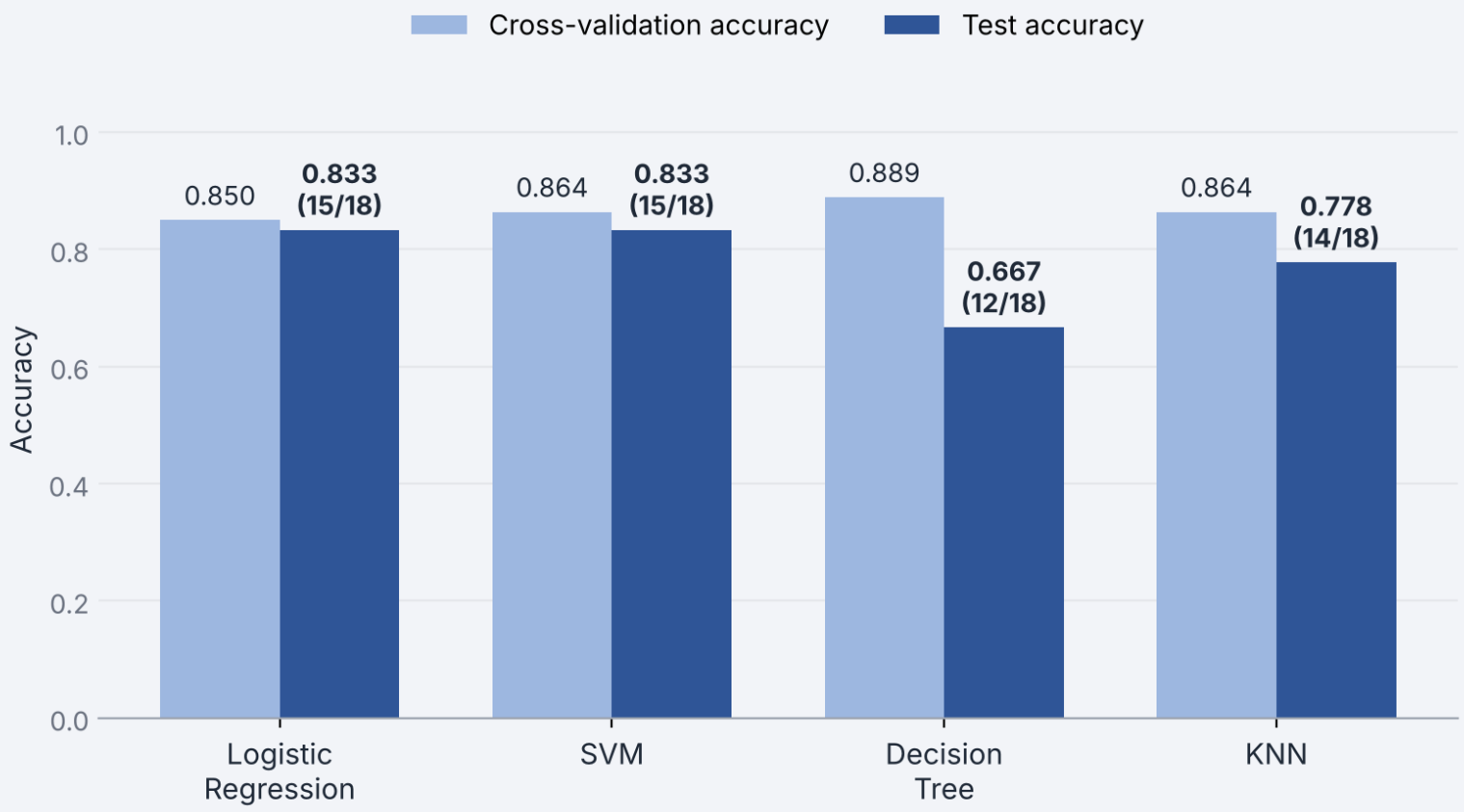
The lowest range had no failed landings, only flights that never tried to land: the same Class = 0 issue as GTO and 2018.

Section 5

Predictive Analysis (Classification)



Logistic Regression and SVM tied for the highest test accuracy



Observations

- Logistic Regression and SVM tied for the highest test accuracy (0.833, 15 of 18).
- Decision Tree had the highest cross-validation accuracy (0.889) but the lowest test accuracy (0.667, 12 of 18).

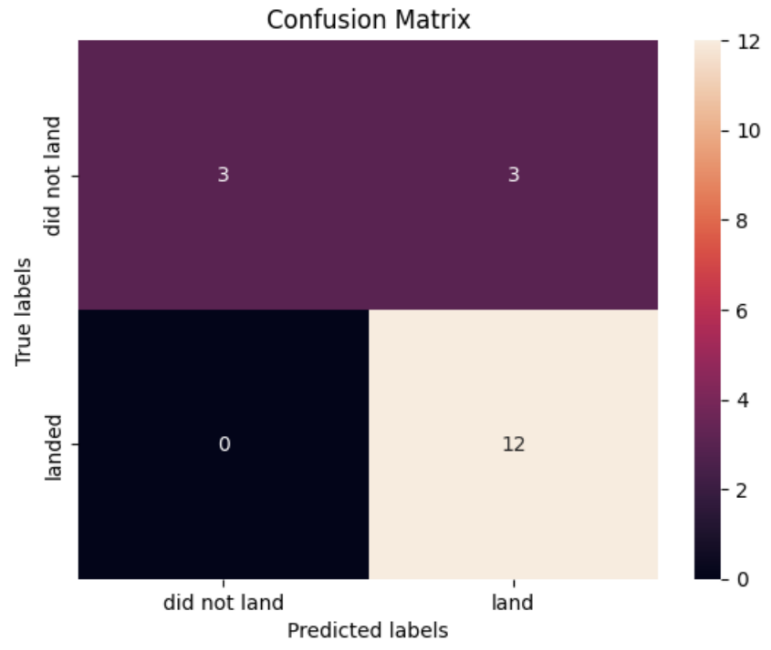
Insight

0.833 — **Logistic Regression and SVM (tied)**
15 of 18 test flights

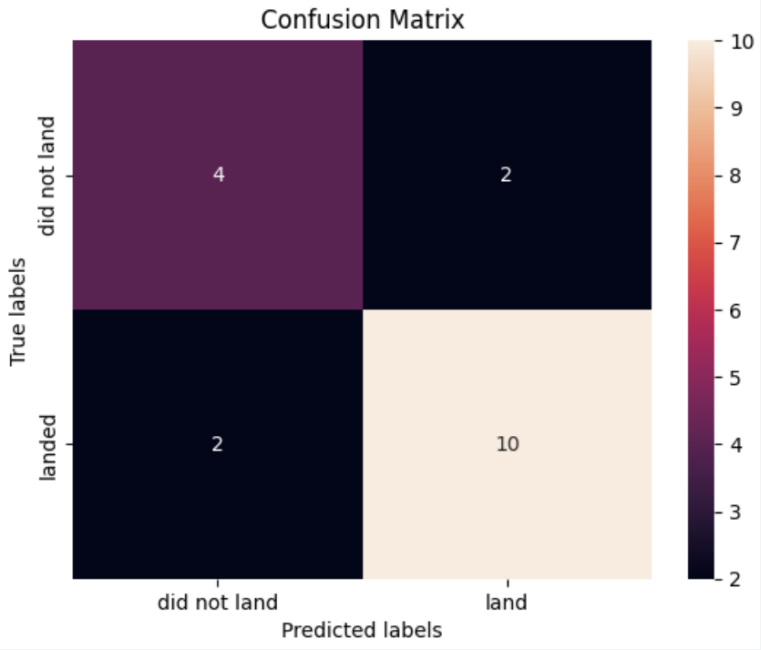
5.6 pt — **One test flight**
the top models differ by only 1–2 flights

With only 18 test flights, the ranking is not very reliable.
Cross-validation and test results should be read together.

Logistic Regression and SVM tend to predict “land”; KNN catches more failures



Logistic Regression / SVM (identical)
15 of 18 correct: 3 false positives, 0 false negatives



KNN
14 of 18 correct: 2 false positives, 2 false negatives

Observations

- Logistic Regression and SVM produced the same confusion matrix: all 12 landings were predicted correctly, but 3 of 6 failures were predicted as “land” (false positives).
- KNN caught one more failure (4 of 6) but missed 2 landings (false negatives).

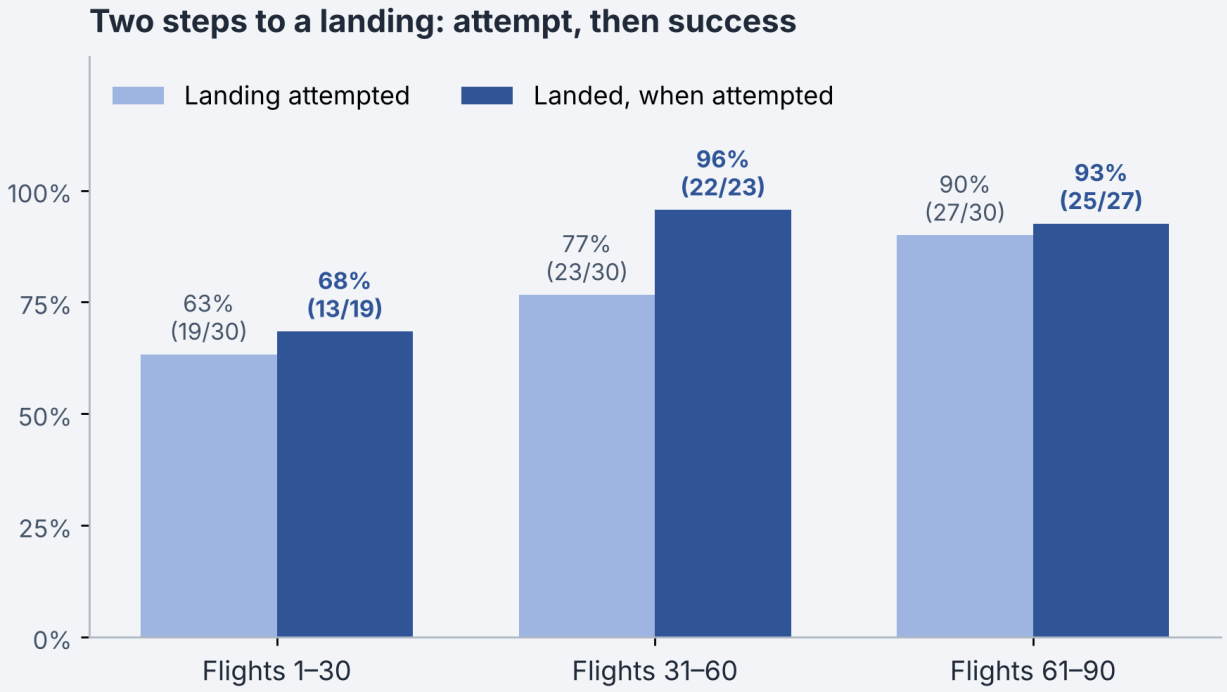
Insight

3/6 **Failures caught: LR / SVM**
all 12 landings caught

4/6 **Failures caught: KNN**
10 of 12 landings caught

The best model depends on which error costs more (a false positive underestimates cost). 18 test flights.

The real question is whether SpaceX tries to land, not whether it can



Insight

93–96%

Landed, when attempted
flights 31–90 (47 of 50)

12 / 21

No-attempt flights were GTO
the other 9 were spread across 5 orbits

For bidding, first ask whether SpaceX plans to recover the booster.

Next step: a two-stage model (attempt, then landing).

Four data limits narrow what the results can claim

1 Mixed label

Class = 0 means two different things

- 21 of 30 Class 0 flights made no landing attempt
- GTO: 52% overall, but 93% (14 of 15) when a landing was attempted
- 5 ocean landings count as success, but those boosters were not recovered

Impact: The model partly predicts whether SpaceX tries to land, not only whether the landing works

2 Small sample

90 flights, 18 in the test set

- One test flight = 5.6 percentage points
- Top models differ by only 1–2 test flights
- 95% interval for 0.833: about 0.67–1.00 (bootstrap), overlapping the 0.667 baseline

Impact: The model ranking is not reliable; read cross-validation and test results together

3 Overlapping factors

Time and hardware moved together

- Flight Number and Block: correlation 0.93
- Site, orbit and payload gaps mostly disappear when flights from the same period are compared

Impact: We cannot separate experience from booster upgrades, or claim causes

4 Inconsistent sources

Different row counts and labels

- Rows: SQL 101, API 90, Folium/Dash 56
- Labels: trailing spaces, CCAFS LC-40 vs SLC-40, LEO vs VLEO
- Periods: SQL to Dec 2020, API to Nov 2020, Folium/Dash to Jun 2018

Impact: Numbers depend on the source, and may not reflect Falcon 9 today

Conclusions: landings are predictable, and timing matters most

Main conclusion

Pre-launch data can predict Falcon 9 landings (15 of 18), and timing explains most of the differences.

43% → 83%

Success rate rose with experience

Flights 1–30 vs 61–90 (13/30 → 25/30)

Mostly timing

Site, orbit and payload gaps shrink

e.g. KSC LC 39A 0.77 vs others 0.79 in the same period

47 / 50

Attempted landings succeeded

since flight 31; 12 of 21 no-attempt flights were GTO

0.833 vs 0.667

Best models vs baseline

LR and SVM; errors are false positives; only 18 test flights

Recommendation

When bidding, first check whether SpaceX plans to recover the booster, then use the model as a guide.

Appendix

Repository: https://github.com/hokuamadata/spacex-falcon9-landing-prediction

01 Data collection (API) nbviewer

02 Web scraping nbviewer

03 Data wrangling nbviewer

04 EDA with SQL nbviewer

05 EDA with visualization nbviewer

06 Interactive map (Folium) nbviewer

07 Dashboard (Dash)
app files (GitHub) · notebook (nbviewer)

08 Machine learning prediction nbviewer

Appendix

Source	Rows	Period	Used in
SpaceX REST API (Falcon 9 only)	**90**	Jun 2010 – Nov 2020	Data collection, wrangling, EDA charts, ML
SQL table (Spacex.csv)	**101**	Jun 2010 – Dec 2020	EDA with SQL
Folium / Dash CSV	**56**	Jun 2010 – Jun 2018	Interactive map, dashboard

Appendix : Evidence for data limits

```
Outcome
False ASDS     6   0
False Ocean    2   0
False RTLS     1   0
None ASDS      2   0
None None     19   0
True ASDS      0  41
True Ocean     0   5
True RTLS      0  14
Class 0 flights with no landing attempt: 21 / 30
GTO all flights: 52% (14/27)
GTO attempted only: 93% (14/15)
Ocean landings counted as success: 5
```

Supports: Data limits card 1, Executive Summary

```
Correlation of FlightNumber and Block: 0.93

KSC LC 39A vs others (same period = flight 27 onward)
           all flights      same period
                  mean size        mean size
LaunchSite
False             0.63   68        0.79   42
True              0.77   22        0.77   22

Payload >= 10,000 kg vs lighter (same period = flight 63 onward)
            all flights      same period
                   mean size        mean size
PayloadMass
False              0.63   75        0.85   13
True               0.87   15        0.87   15

VLEO vs other orbits (same period = flight 65 onward)
      all flights      same period
             mean size        mean size
Orbit
False        0.63   76        0.83   12
True         0.86   14        0.86   14
```

Supports: Data limits card 3, Conclusions point 2

Appendix : Evidence for insight and model

Period	flights	attempted	landed	attempt_rate	landed_when_attempted
1–30	30	19	13	0.63	0.68
31–60	30	23	22	0.77	0.96
61–90	30	27	25	0.90	0.93

Orbit
GTO 12
ISS 4
LEO 2
PO 1
MEO 1
SO 1
Name: count, dtype: int64

Supports: Insight slide,
Conclusions points 1 and 3

```
Test flights: 18, one flight = 5.6 percentage points
Baseline (always predict landing): 0.667
Accuracy 0.833, 95% bootstrap interval 0.67–1.00
```

Supports: Data limits card 2,
Conclusions point 4

Thank you!

